

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

December 2012

EE402 Electrical Power System-IV

Date: 04.12.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

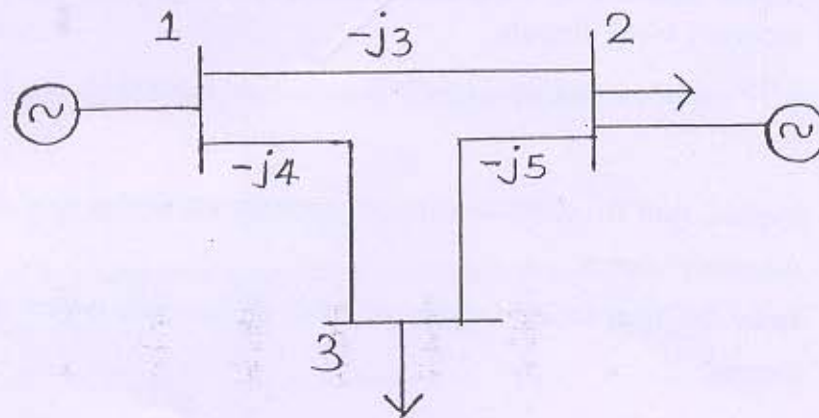
Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Give the flow chart for a load flow study on a power system having both P-Q as well as P-V buses using G-S method. [07]

Q - 2 (a) A 3 bus power system is shown in figure given below. The relevant per unit line admittances on 100MVA base are indicated on diagram and bus data are given in table. [10]



Form Y_{BUS} and determine the voltages at bus 2 and 3 after the second iteration using Gauss-Seidel method. Take the acceleration factor $\alpha=1.6$.

Bus	P_L (MW)	Q_L (MVar)	P_G (MW)	Q_G (MVar)	V (pu)	Remarks
1	0	0	?	?	1.02/_0	Slack bus
2	50	25	25	15	?	PQ bus
3	60	30	0	0	?	PQ bus

(b) What is economic dispatch? Explain Heat rate curve and incremental fuel cost curve of thermal power plant. [04]

OR

Q - 2 (a) The fuel cost of three thermal plants in \$/h are given as:

[07]

$$C_1 = 500 + 5.3P_1 + 0.004P_1^2$$

$$C_2 = 400 + 5.5P_2 + 0.006P_2^2$$

$$C_3 = 200 + 5.8P_3 + 0.009P_3^2$$

Where the powers are in MW. When the total load is 975 MW with the following generator limits (in MW):

$$200 \leq P_1 \leq 450$$

$$150 \leq P_2 \leq 350$$

$$100 \leq P_3 \leq 225$$

Neglecting line losses find the optimal dispatch and the total cost in \$/h by iteration. Assume the initial value of $\lambda^{(1)} = 6$.

(b) Derive the equation of transmission loss formula.

[07]

Q - 3 (a) Obtain closed loop transfer function of Automatic Voltage Regulator loop with necessary block diagram. [10]

(b) Why one of the buses is taken as slack bus in load flow studies?

[04]

OR

Q - 3 (a) Explain load frequency control of generator and derive final block diagram of load frequency control. [10]

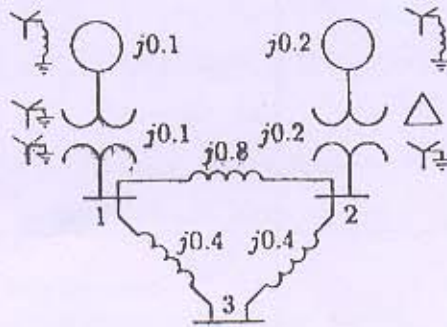
(b) Draw the final block diagram of AGC in two area system including tie line bias control. [04]

SECTION - II

Q - 4 What is short circuit capacity? Derive equation for SCC.

[07]

Q - 5 (a) A single line diagram of a simple 3 bus power system network is shown in figure. All impedances are expressed in per unit on a common 100MVA base and for simplicity resistances are neglected. Shunt capacitances are neglected and the system is considered on no load. All generators are running at their rated voltage and rated frequency with their emfs in phase. Determine the fault currents, the bus voltages and line currents during a fault using thevenin's theorem when a balanced three phase fault with a fault impedance $Z_f = 0.16$ per unit occurs on (a) bus 3 and (b) bus 2. [10]



(b) Give comparison between STATCOM and SVC. [04]

OR

Q - 5 (a) Explain unbalanced fault analysis using bus impedance matrix for single line to ground, line to line and double line to ground fault. [06]

(b) Discuss the phenomenon of wave reflection and refraction. Derive expressions for reflection and refraction coefficient. [08]

Q - 6 (a) What do you mean by compensation? Explain load compensation and system compensation. [07]

(b) How can bewley lattice diagram be drawn? Discuss its use. [07]

OR

Q - 6 (a) What is arcing ground? On which system does it occur? [07]

(b) What type of lightning strokes can occur on transmission line? [07]
